

Joint Trajectory Control, Power Control, and Collection Schedule in UAV-Assisted Anti-Jamming Wireless Data Collection With Imperfect CSI

Yuan Bian^{1b}, Jianbo Hu, Peng Zhang^{1b}, Shuo Wang, Yingyang Wang^{1b}, Jiping Cong, and Chaoqi Fu^{1b}

Abstract—This letter presents a new approach to enhance the anti-jamming capability of unmanned aerial vehicle (UAV)-assisted wireless data collection, specifically addressing the challenge of multi-jammers attacks in the context of imperfect channel state information (CSI). The proposed approach is designed for a scenario where a single UAV is responsible for collecting data from multiple ground sensors, while encountering multi-jammers from a ground-based adversary. The objective is to maximize the average data rate of all sensors while ensuring a sufficient amount of data is collected from each sensor. The optimization problem is formulated as a mixed-integer non-convex problem, and successive convex approximation (SCA) technique is employed, which jointly optimizes the collection schedule, power control and UAV trajectory. Numerical simulations demonstrate that the proposed approach outperforms in the presence of interference and imperfect CSI.

Index Terms—UAV, trajectory control, average rate, power control, collection schedule.

I. INTRODUCTION

UAV-ASSISTED wireless data collection has significantly enhanced convenience across various application domains, including environmental monitoring, disaster management, and surveillance [1], [2], [3]. However, the open nature of wireless communication channels exposes the connections between sensors and UAVs to significant vulnerability from interference attacks. Therefore, ensuring the feasibility and effectiveness of UAV communication system under malicious jamming attacks remains a significant challenge [4].

The existing literature has extensively explored the challenges associated with anti-jamming UAV communication scenarios, encompassing A2G (air-to-ground) data transmission [5], mobile edge computing [6] relay communication [7] and G2A (ground-to-air) data collection [8], [9], [10], among others. Reference [8] introduced a trajectory planning approach that optimizes the UAV's three-dimensional trajectory using successive convex approximation (SCA), where the UAV communicates with only one ground node. Specifically in [9], the ground sensors (GSs) transmission scheduling and UAV's three-dimensional trajectory under the probabilistic

line-of-sight (LoS) channel model are jointly optimized to maximize the minimum (average) expected rate. Additionally, a trajectory optimization method was proposed to anti jamming attacks of UAV, leveraging a dueling double deep Q-network (D3QN) [10].

With regard to the anti-jamming scheme, previous work primarily focused on UAV communication with only one ground node or only one jammer; some works assume that the perfect CSI between the UAV and the jammer when optimizing the UAV's trajectory, neglecting the uncertainty of the channel state. Therefore, conducting research on UAV-assisted anti-jamming data collection of sensors with imperfect CSI under the condition of multi-jammers is challenging.

Motivated by the concerns outlined above, this letter introduces an optimization problem aimed at maximizing the average data rate in the presence of CSI and multi-jammers, while adhering to schedule, power and motion constraints. Specifically, a worst-case criterion is integrated with the channel model to address the uncertainty associated with CSI. An alternating optimization strategy is then employed, where the collection schedule, power control and UAV trajectory are iteratively optimized while keeping the other variable fixed, until the convergence is achieved. The iterative SCA approach is utilized to solve the two non-convex optimization subproblems. Finally, numerical results are presented to validate the theoretical analysis.

II. SYSTEM MODEL AND PROBLEM FORMATION

We consider a wireless sensor network (WSN) situated in a suburban area, comprising a set of $\mathcal{S}$ sensor nodes on the ground that transmit information to a UAV as Fig. 1 shows. The sensor coordinate is denoted as $\mathbf{q}_s[n] = (x_s, y_s)$, $s \in \mathcal{S}$. Meanwhile, a jammer on the ground sends signals to interfere with the sensor-UAV link. It is assumed that the collection duration, known as the flight time, is fixed at T and divided into N equal time slots, indexed by $n = 1, \dots, N$. And the duration of each time slot can be expressed as δ_t .

Similarly, the 2-D plane coordinates location of the jammer is expressed as $\mathbf{q}_j = (x_j, y_j)$, $j \in \mathcal{J}$, which remains constant over a time slot. The UAV's 2-D plane coordinates at slot n is given by $\mathbf{q}_u[n] = (x_u[n], y_u[n])$, while the height H is supposed to be a constant. It is assumed that δ_t is sufficiently small that the speed of UAV is unchanged within each time slot even at the maximum flying speed.

The speed of the UAV at time slot n $\mathbf{v}[n]$ is constrained within a specific range $[v_{\min}, v_{\max}]$, which in turn limits the distance it could cover. Additionally, there exists an upper limit on the acceleration $\mathbf{a}[n]$ of the UAV represented as $a_{\max}$. The followings are the relevant constraints:

$$\|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\|^2 \leq v_{\max}^2 \times \delta_t^2, \quad n = 1, \dots, N \quad (1)$$

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Yuan Bian, Jianbo Hu, Peng Zhang, Yingyang Wang, Jiping Cong, and Chaoqi Fu are with the School of Equipment Management and UAV Engineering, Air Force Engineering University, Xi'an 710071, China, and also with the Youth Innovation Team, Shanxi University, Xi'an 710071, China (e-mail: bianyuanlara@163.com; jian_bo_h@163.com; wryuofu@163.com; wangyingyang@outlook.com; bugu1233@163.com; fuchaoqi2011@163.com).

Shuo Wang is with the State Key Laboratory of Astronautic Dynamics, Xi'an Satellite Control Center, Xi'an 710043, China, and also with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: shuowang@xidian.edu.cn).

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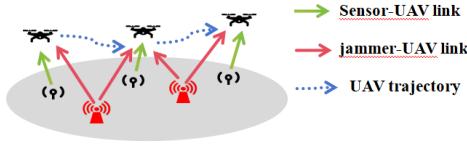


Fig. 1. Illustration of the proposed model.

$$\|\mathbf{v}[n]\| \leq v_{\max}, \quad n = 1, \dots, N \quad (2)$$

$$\|\mathbf{v}[n]\| \geq v_{\min}, \quad n = 1, \dots, N \quad (3)$$

$$\|\mathbf{a}[n]\| \leq a_{\max}, \quad n = 1, \dots, N \quad (4)$$

$$\mathbf{v}[0] = \mathbf{v}_0, \quad \mathbf{v}[N] = \mathbf{v}_N \quad (5)$$

$$\mathbf{q}_u[0] = \mathbf{q}_0, \quad \mathbf{q}_u[N] = \mathbf{q}_N \quad (6)$$

$$\mathbf{v}[n] = \mathbf{v}[n-1] + \mathbf{a}[n]\delta_t, \quad n = 2, \dots, N \quad (7)$$

$$\mathbf{q}_u[n] = \mathbf{q}_u[n-1] + \mathbf{v}[n]\delta_t + \frac{1}{2}\mathbf{a}[n]\delta_t^2 \quad (8)$$

where $\mathbf{v}_0$ is the initial velocity of UAV and $\mathbf{v}_N$ represents the final velocity; $\mathbf{q}_0$ is the initial location of the UAV and $\mathbf{q}_N$ is the final location.

Assume that the sensor-UAV link is primarily characterized by LoS connection [11], [12], then the channel power gain from sensor s to UAV u during slot n adopts the free-space path loss model, which can be expressed as:

$$h_{su}[n] = \beta_0 d_{su}[n]^{-2} = \frac{\beta_0}{(\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2 + H^2)} \quad (9)$$

where β_0 represents the channel power gain at the reference distance $d_0 = 1\text{m}$, $d_{su}[n]$ denotes the distance from sensor s to UAV u at slot n . Similarly, the channel power gain from jammer to UAV at slot n is

$$h_{ju}[n] = \beta_0 d_{ju}[n]^{-2} = \frac{\beta_0}{(\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2 + H^2)} \quad (10)$$

Hence, the transmission rate $R_{su}[n]$ at which the sensor s uploads the perceptual information to the UAV is expressed as follows.

$$R_{su}[n] = \log_2 \left(1 + \frac{P_s[n]h_{su}[n]}{\sum_{j=1}^J P_j h_{ju}[n] + \sigma^2} \right) \quad (11)$$

where the transmitting power of the sensor and jammer are represented by P_s and P_j , respectively. The constraints pertaining to the range and average transmit power of the sensor are outlined below. σ^2 denotes the noise power.

$$0 \leq P_s[n] \leq 1, \quad \frac{1}{N} \sum_{n=1}^N P_s[n] \leq 0.9. \quad (12)$$

Define a binary variable $\varepsilon[n]$, which indicates that the data is uploaded from sensor s to UAV u at time slot n . Note that in each time slot, only one sensor's data is collected by the UAV. Therefore, the following constraints must be met.

$$\varepsilon_s[n] \in \{0, 1\}, \quad \sum_{s=1}^S \varepsilon_s[n] \leq 1 \quad (13)$$

Due to perception errors, quantization errors, and other factors leading to incomplete information obtained by the UAV

regarding the sensor-UAV and jammer-UAV links, the accurate acquisition of CSI for these links is hindered.

In fact, there are two types of CSI error models: the bounded CSI error model and the statistical CSI error model [13]. In this letter, the bounded CSI error model is employed to characterize the errors present in the direct channel. Consequently, a bounded channel state information uncertainty model is employed to describe the channel state as below.

$$\Omega_s \in \{\Delta h_{su} | h_{su} = \bar{h}_{su} + \Delta h_{su} : |\Delta h_{su}| \leq \psi_s \leq \bar{h}_{su}\} \quad (14)$$

$$\Omega_j \in \{\Delta h_{ju} | h_{ju} = \bar{h}_{ju} + \Delta h_{ju} : |\Delta h_{ju}| \leq \psi_j \leq \bar{h}_{ju}\} \quad (15)$$

where $\bar{h}_{su}$ and $\bar{h}_{ju}$ represent the channel estimation values, Δh_{su} and Δh_{ju} denote the channel estimation errors. $\psi_s \geq 0$ and $\psi_j \geq 0$ represent the upper bounds of the CSI. Ω_s and Ω_j represent channel gain uncertainty set.

In this letter, we aim to maximize the average rate of sensor-UAV link while mitigating the effects of malicious jamming, by jointly optimizing the collection schedule $\{\varepsilon_s[n]\}$, sensor power control $\{P_s[n]\}$ and UAV trajectory $\{\mathbf{q}_u[n]\}$. The problem can be formulated as

P(1.1) :

$$\begin{aligned} \max_{\{\mathbf{q}_u[n], \varepsilon_s[n], P_s[n]\}} & \frac{1}{N} \sum_{n=1}^N \sum_{s=1}^S \varepsilon_s[n] R_{su}[n] (\Delta h_{su}, \Delta h_{ju}) \\ \text{s.t.} & \begin{cases} B \sum_{n=1}^N R_{su}[n] (\Delta h_{su}, \Delta h_{ju}) \geq M & (16-1) \\ \Delta h_{su} \in \Omega_s, \quad \Delta h_{ju} \in \Omega_j & (16-2) \\ (1)-(8), (12), (13) \end{cases} \end{aligned} \quad (16)$$

where

$$R_{su}[n] = \log_2 \left(1 + \frac{P_s[n](\bar{h}_{su}[n] + \Delta h_{su})}{\sum_{j=1}^J P_j (\bar{h}_{ju}[n] + \Delta h_{ju}) + \sigma^2} \right) \quad (17)$$

Note that (1)-(8) represent the UAV's trajectory constraints, (12) indicates the power constraint, and (13) defines the sensor-UAV schedule. Constraint (16-1) ensures that the data uploaded from each sensor to the UAV meets the specified requirements, where M denotes the minimum amount of data to be uploaded and B represents the bandwidth.

III. PROPOSED ALGORITHM

Given the uncertainty of parameters in problem P(1.1), it is initially transformed into a deterministic problem based on the worst-case criterion [14]. Subsequently, the optimization process proceeds by fixing one variable while optimizing the others, alternating between these two sub-problems iteratively until convergence is achieved.

A. Uncertainty Transformation Of Objection Function (16)

To transform the uncertainty of parameters in objective function into a deterministic form, the following should be

satisfied according to the worst-case criterion.

$$\begin{aligned} & \max_{\Delta h_{su}, \Delta h_{ju}} \frac{1}{N} \sum_{n=1}^N \sum_{s=1}^S \varepsilon_s[n] R_{su}[n] (\Delta h_{su}, \Delta h_{ju}) \\ & \Rightarrow \log_2 \left(1 + \frac{\min_{\Delta h_{su}} (P_s[n] (\bar{h}_{su}[n] + \Delta h_{su}))}{\max_{\Delta h_{ju}} (\sum_{j=1}^J P_j (\bar{h}_{ju}[n] + \Delta h_{ju})) + \sigma^2} \right) \\ & \Rightarrow \log_2 \left(1 + \frac{P_s (\bar{h}_{su}[n] - \psi_s)}{\sum_{j=1}^J P_j (\bar{h}_{ju}[n] + \psi_j) + \sigma^2} \right) \end{aligned} \quad (18)$$

Given the above transformation, the problem P(1.1) could be reformulated as

P(1.2) :

$$\begin{aligned} & \max_{\{\mathbf{q}_u[n], \varepsilon_s[n], P_s[n]\}} \frac{1}{N} \sum_{n=1}^N \sum_{s=1}^S \varepsilon_s[n] \tilde{R}_{su}[n] \\ & \text{s.t.} \begin{cases} B \sum_{n=1}^N \varepsilon_s[n] \tilde{R}_{su}[n] \geq M, & s \in \mathcal{S} \quad (19-1) \\ (1) - (8), (12), (13) \end{cases} \end{aligned} \quad (19)$$

where

$$\tilde{R}_{su}[n] = \log_2 \left(1 + \frac{P_s[n] (\bar{h}_{su}[n] - \psi_s)}{\sum_{j=1}^J P_j (\bar{h}_{ju}[n] + \psi_j) + \sigma^2} \right) \quad (20)$$

Note that problem P(1.2) is a non-convex optimization problem due to non-concave objective function and non-convex constraints. To tackle this issue, the problem is decomposed into two sub-problems: joint collection schedule and power control, along with UAV trajectory optimization.

B. Joint Collection Schedule And Power Optimization

In the following, the non-concave objective function (19), constraints (13) and (19-1) are transformed to achieve convexity. This process begins by relaxing the binary variable $\varepsilon_s[n]$ into a continuous variable that ranges from 0 to 1, followed by replacing the constraints with

$$0 \leq \varepsilon_s[n] \leq 1, \quad \sum_{s=1}^S \varepsilon_s[n] \leq 1 \quad (21)$$

To resolve the non-concave objective function (19), a slack variable $\{y[n]\}$ is introduced, leading to the transformation of problem P(2.1).

$$\begin{aligned} & \text{P(2.1)} : \max_{\{\varepsilon_s[n], P_s[n]\}} \frac{1}{N} \sum_{n=1}^N \sum_{s=1}^S \varepsilon_s[n] y[n] \\ & \text{s.t.} \begin{cases} y[n] \leq \tilde{R}_{su}[n] \quad (22-1) \\ B \sum_{n=1}^N \varepsilon_s[n] y[n] \geq M, & s \in \mathcal{S} \quad (22-2) \\ (1) - (8), (12), (21) \end{cases} \end{aligned} \quad (22)$$

Note that the problem P(2.1) above is still non-convex due to the coupled variables in objective function and constraint (22-2). To this end, $\varepsilon_s[n] y[n]$ is transformed

$$\varepsilon_s[n] y[n] = \frac{(\varepsilon_s[n] + y[n])^2 - \varepsilon_s[n]^2 - y[n]^2}{2} \quad (23)$$

Then, a local convex approximation is applied. Specifically, since $(\varepsilon_s[n] + y[n])^2$ is convex and differentiable with respect to $\varepsilon_s[n]$ and $y[n]$, we have

$$\begin{aligned} (\varepsilon_s[n] + y[n])^2 & \geq (\varepsilon_s^r[n] + y^r[n])^2 \\ & \quad + 2(\varepsilon_s^r[n] + y^r[n])(\varepsilon_s[n] - \varepsilon_s^r[n]) \\ & \quad + 2(\varepsilon_s^r[n] + y^r[n])(y[n] - y^r[n]) \\ & \triangleq x_r^{lb}[n] \end{aligned} \quad (24)$$

Therefore, the problem is approximated as

P(2.2) :

$$\begin{aligned} & \max_{\{\varepsilon_s[n], P_s[n]\}} \frac{1}{N} \sum_{n=1}^N \sum_{s=1}^S \frac{(x_r^{lb}[n] - \varepsilon_s[n]^2 - y[n]^2)}{2} \\ & \text{s.t.} \begin{cases} y[n] \geq \frac{(x_r^{lb}[n] - \varepsilon_s[n]^2 - y[n]^2)}{2} \\ B \sum_{n=1}^N \frac{(x_r^{lb}[n] - \varepsilon_s[n]^2 - y[n]^2)}{2} \geq M, \\ s \in \mathcal{S} \quad (25-1) \\ (1) - (8), (12), (21), (22-1) \end{cases} \end{aligned} \quad (25)$$

Since the problem P(2.2) is convex, it can be effectively solved using CVX.

C. UAV Trajectory Optimization

Furthermore, given a specified collection schedule and power $\{\varepsilon_s[n], P_s[n]\}$, the optimization of the UAV trajectory $\{\mathbf{q}_u[n]\}$ aims to maximize the average rate of all sensors. It could be precisely formulated as:

$$\begin{aligned} & \text{P(3.1)} : \max_{\{\mathbf{q}_u[n]\}} \frac{1}{N} \sum_{n=1}^N \sum_{s=1}^S \varepsilon_s[n] \tilde{R}_{su}[n] \\ & \text{s.t.} \quad (1-8), (19-1) \end{aligned} \quad (26)$$

Note that P(3.1) is a non-convex optimization problem due to the presence of non-concave objective function and non-convex constraints (3) and (19-1). To tackle this problem, $\tilde{R}_{su}[n]$ can be expressed as:

$$\tilde{R}_{su}[n] = \hat{R}_{su}[n] - \hat{R}_{su}[n] \quad (27)$$

with

$$\begin{aligned} \hat{R}_{su}[n] & = \log_2 \left(1 + \frac{P_s[n] \beta_0}{(\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2 + H^2) \sigma^2} - \frac{P_s[n] \psi_s}{\sigma^2} \right. \\ & \quad \left. + \sum_{j=1}^J \frac{P_j \beta_0}{(\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2 + H^2) \sigma^2} + \frac{P_j \psi_j}{\sigma^2} \right) \end{aligned} \quad (28)$$

$$\hat{R}_{su}[n] = \log_2 \left(\frac{P_j \beta_0}{(\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2 + H^2) \sigma^2} + \frac{P_j \psi_j}{\sigma^2} + 1 \right) \quad (29)$$

Since $\hat{R}_{su}[n]$ is convex with respect to $\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2$ and $\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2$, and with known $\|\mathbf{q}_j - \mathbf{q}_u^r[n]\|^2$ and $\|\mathbf{q}_s - \mathbf{q}_u^r[n]\|^2$, the following lower bound for $\hat{R}_{su}[n]$ at the r -th iteration is obtained by applying the first-order Taylor Expansion.

$$\begin{aligned} \hat{R}_{su}[n] &\geq \log_2(G[n]) \\ &\quad - \sum_{j=1}^J A[n](\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2 - \|\mathbf{q}_j - \mathbf{q}_u^r[n]\|^2) \\ &\quad - I[n](\|\mathbf{q}_s - \mathbf{q}_u[n]\|^2 - \|\mathbf{q}_s - \mathbf{q}_u^r[n]\|^2) \\ &\triangleq \phi(\hat{R}_{su}[n]) \end{aligned} \quad (30)$$

with

$$\begin{aligned} G[n] &= 1 + \frac{P_s[n]\beta_0}{(\|\mathbf{q}_s - \mathbf{q}_u^r[n]\|^2 + H^2)\sigma^2} - \frac{P_s[n]\psi_s}{\sigma^2} \\ &\quad + \sum_{j=1}^J \left(\frac{P_j\beta_0}{(\|\mathbf{q}_j - \mathbf{q}_u^r[n]\|^2 + H^2)\sigma^2} + \frac{P_j\psi_j}{\sigma^2} \right) \end{aligned} \quad (31)$$

$$A[n] = \frac{1}{G[n]} \frac{P_j\beta_0}{(\|\mathbf{q}_j - \mathbf{q}_u^r[n]\|^2 + H^2)^2\sigma^2} \quad (32)$$

$$I[n] = \frac{1}{G[n]} \frac{P_s[n]\beta_0}{(\|\mathbf{q}_s - \mathbf{q}_u^r[n]\|^2 + H^2)^2\sigma^2} \quad (33)$$

Note that is $\hat{R}_{su}[n]$ convex with respect to $\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2$, the slackening variable $\{z_j[n]\}$ is introduced to satisfy the following inequality constraint.

$$\frac{(\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2 + H^2)\sigma^2}{P_j\beta_0} \geq e^{z_j[n]} \quad (34)$$

Since $\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2$ is convex and differentiable with respect to $\mathbf{q}_u[n]$, its lower bound at the r -th iteration is obtained by applying the first-order Taylor Expansion.

$$\begin{aligned} \|\mathbf{q}_j - \mathbf{q}_u[n]\|^2 &\geq \|\mathbf{q}_j - \mathbf{q}_u^r[n]\|^2 \\ &\quad + 2(\mathbf{q}_j - \mathbf{q}_u^r[n])^T(\mathbf{q}_u^r[n] - \mathbf{q}_u[n]) \\ &\triangleq \phi(\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2) \end{aligned} \quad (35)$$

Hence,

$$e^{z_j[n]} \leq \frac{(\phi(\|\mathbf{q}_j - \mathbf{q}_u[n]\|^2) + H^2)\sigma^2}{P_j\beta_0} \quad (36)$$

As a result, the subsequent in-equations can be derived

$$\begin{aligned} \tilde{R}_{su}[n] &\geq \phi(\hat{R}_{su}[n]) - \log_2\left(1 + \sum_{j=1}^J (e^{-z_j[n]} + \frac{P_j\psi_j}{\sigma^2})\right) \\ &\triangleq \phi(\tilde{R}_{su}^{lb}[n]) \end{aligned} \quad (37)$$

Then, constraint (19-1) is transformed as follows:

$$B \sum_{n=1}^N \varepsilon_s[n] \phi(\tilde{R}_{su}^{lb}[n]) \geq M, \quad s \in \mathcal{S} \quad (38)$$

Note that constraint (3) is non-convex, it could be transformed into the follow in-equation.

$$\|\mathbf{v}[n]\|^2 \geq \|\mathbf{v}_0[n]\|^2 + 2\mathbf{v}_0^T[n](\mathbf{v}[n] - \mathbf{v}_0[n]) \geq v_{min}^2 \quad (39)$$

Consequently, problem P(3.1) could be reformulated as a convex optimization problem P(3.2) and be solved using the

Algorithm 1 SCA Algorithm for Solving P(1.2)

1. INITIALIZE($W_s^0[n] = \{\varepsilon_s[n], P_s[n]\}$, $Q_u^0[n] = \{\mathbf{q}_u[n]\}$) and $k = 0$.

2. Repeat

3. With the given Q_u^{k-1} , problem P(2.2) is solved by SCA and $W_s^k = \{\varepsilon_s[n]^*, P_s[n]^*\}$ is obtained.

4. With the given W_s^k , problem P(3.2) is solved and $Q_u^k = \{\mathbf{q}_u[n]^*\}$ is obtained.

5. Updater the variables

$\{\mathbf{q}_u[n], \varepsilon_s[n], P_s[n]\}^k \leftarrow \{\mathbf{q}_u[n], \varepsilon_s[n], P_s[n]\}^*$

6. Updater k

$k = k + 1$

7. Until some termination conditions are met.

CVX solver. The procedure for solving P(2.2) and P(3.2) is outlined below.

$$\begin{aligned} \mathbf{P}(3.2) : \max_{\{\mathbf{q}_u[n]\}} \quad & \frac{1}{N} \sum_{n=1}^N \sum_{s=1}^S \varepsilon_s[n] \phi(\tilde{R}_{su}^{lb}[n]) \\ \text{s.t.} \quad & (1) - (2), (4 - 8), (35) - (40) \end{aligned} \quad (40)$$

D. Algorithm Analysis

In this section, an overview of the convergence property of Algorithm 1 is provided [15]. For the sake of clarity, the objective function value of problem (P1.2), (P2.2) and (P3.2) of r -th iteration are denoted as $R\{W_s, Q_u\}$, $R_s\{W_s, Q_u\}$ and $R_u\{W_s, Q_u\}$, respectively. Then it follows that

$$\begin{aligned} R\{W_s, Q_u^r\} &\stackrel{a}{=} R_s^{lb}\{W_s, Q_u^r\} \\ &\stackrel{b}{\leq} R_s^{lb}\{W_s^{r+1}, Q_u^r\} \stackrel{c}{\leq} R\{W_s^{r+1}, Q_u^r\} \end{aligned} \quad (41)$$

where (a) is valid since the problem (2.2) at $W_s^r = \{\varepsilon_s[n]^*, P_s[n]^r\}$ has the same objective value as that of problem (1.2); (b) holds due to the fact that with the given $Q_u^r = \{\mathbf{q}_u[n]^r\}$, problem (2.2) is solved optimally yielding the optimal solution W_s^{r+1} ; (c) holds since with given Q_u^r , $R_s^{lb}\{W_s^{r+1}, Q_u^r\}$ is the lower bound of $R\{W_s^{r+1}, Q_u^r\}$.

Similarly, an analysis of Step 4 indicates that

$$\begin{aligned} R\{W_s^{r+1}, Q_u^r\} &\stackrel{a}{=} R_u^{lb}\{W_s^{r+1}, Q_u^r\} \\ &\stackrel{b}{\leq} R_u^{lb}\{W_s^{r+1}, Q_u^{r+1}\} \stackrel{c}{\leq} R\{W_s^{r+1}, Q_u^{r+1}\} \end{aligned} \quad (42)$$

Therefore, Algorithm 1 is guaranteed to converge.

$$R\{W_s, Q_u^r\} \leq R\{W_s^{r+1}, Q_u^r\} \leq R\{W_s^{r+1}, Q_u^{r+1}\} \quad (43)$$

The overall complexity of Algorithm 1 is comprehensively characterized by the individual complexities of its constituent steps, including the significant contribution from Step 3 and Step 4. Notably, Step 3, which tackles sub-problem (2.2) via the interior point method denoted as $O(L_1 * (3NS)^{3.5})$, where N and S are the numbers of time slots and sensors, and L_1 (the iterations required for updating the communication schedule and power). Similarly, Step 4, responsible for solving (3.2), contributes a complexity based on L_2 (the

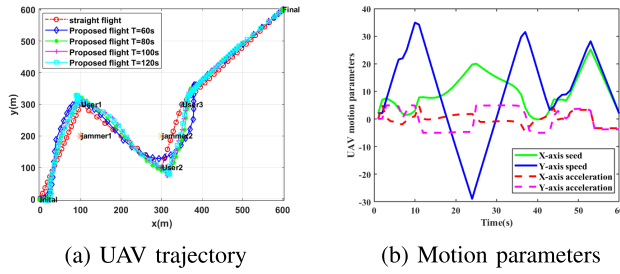


Fig. 2. UAV trajectory and motion parameters.

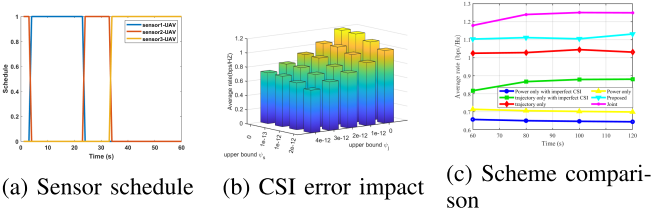


Fig. 3. Schedule, CSI error upper bound impact and average rates of different algorithms.

iterations for updating the UAV trajectory), represented as $O(L_2 * (4NS)^{3.5})$. Consequently, the overall complexity of Algorithm 1 is $O(L_3(L_1 * (3NS)^{3.5} + L_2 * (4NS)^{3.5}))$, encompassing L_3 iterations for convergence.

IV. SIMULATION RESULTS AND ANALYSIS

In this section, we present simulation results to validate the performance of the proposed design. The scenario assumes the deployment of UAV for data collection within a geographical area of $0.6 \times 0.6 \text{ km}^2$. The sensors are specifically located at $[400,200]\text{m}$, $[200,300]\text{m}$, and $[350,400]\text{m}$, while the jammers are positioned at $[100,200]\text{m}$ and $[300,200]\text{m}$. The simulation parameters used are set as follows: $\sigma^2 = -110\text{dBm}$, $\beta_0 = -50\text{dB}$, $v_{\min} = 2\text{m/s}$, $a_{\max} = 5\text{m/s}^2$, $H = 100\text{m}$, $B = 1\text{MHz}$, $M = 10\text{Mbit}$, $P_j = 1.5\text{W}$.

Fig. 2(a) illustrates that under different duration T , the UAV gradually tends to change the trajectory to approach the sensor and move away from the jammers to achieve an optimal rate with $\psi_s = 10^{-12}$, $\psi_j = 10^{-12}$. Fig. 2(b) illustrates the changes in the UAV's speed and acceleration over time. The abrupt changes in speed along the Y-axis are due to alterations in the direction of the UAV's flight trajectory, while the sudden change in speed along the X-axis occurs because the UAV reaches the endpoint with a speed of 0.

Fig. 3(a) shows the sensors-UAV schedule with duration $T = 60\text{s}$, $\psi_s = 10^{-12}$, $\psi_j = 10^{-12}$. It can be concluded that the sensors are scheduled to communicate with the UAV sequentially. Specifically, sensor 1 primarily uploads its data during the initial period, sensor 2 during the middle period, and sensor 3 towards the end of the duration. This scheduling pattern is generally consistent with their respective positions. Fig. 3(b) shows average rate with duration $T = 60\text{s}$ versus different channel error upper bound. It can be observed that the rate decreases with channel error upper bound increasing.

In Fig. 3(c), to demonstrate the performance of the proposed scheme, we compare it with five other approaches: (1) Power

only with imperfect CSI; (2) Trajectory only with imperfect CSI; (3) Power only with perfect CSI; (4) Trajectory only with perfect CSI; and (5) Joint with perfect CSI. Furthermore, it can be found that the algorithm we proposed superior to the benchmark algorithms of (1)-(4) under the corresponding settings with duration $T = 60\text{s}$, $\psi_s = 10^{-12}$, $\psi_j = 10^{-12}$.

V. CONCLUSION

In this letter, we optimized the average rate of sensor-UAV while ensuring sufficient data collection in the presence of jamming and imperfect CSI. The optimization problem is first decomposed into two sub-optimization problems: the optimization of the data collection schedule and power control with fixed UAV trajectory and the optimization of the UAV trajectory with fixed collection schedule and power control, which are performed alternately using the SCA method. Finally, the simulation and numerical results verified the proposed scheme.

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